NEUTRONS FOR MEMBRANES: OPPORTUNITIES AND ACCESS AT THE ILL

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ABSTRACT

In this short presentation I will introduce the range of neutron techniques available at the ILL for membrane science, including diffraction, small-angle neutron scattering (SANS), reflectometry, and spectroscopy. I will then illustrate the potential of membrane diffraction, a method dedicated to membrane structural studies, that has been pioneered at Brookhaven and at the ILL on the D16 instrument [1] in the 1970s, and has been continuously improved ever since. The method relies on the combination of neutron diffraction by solid supported, oriented lamellar stacks of membranes with precise control of temperature and humidity [2]. I will illustrate the results obtained with this technique with a selection of past studies from the ILL user program and from my own research. These span from simple synthetic models [3], skin lipids [4-7], plant lipids from thylakoids [9-11], to studies of the myelin structure [12-13]. Finally, I will outline the proposal process to apply for beamtime, the support available to first-time users, with the aim of lowering the barrier to entry for the membrane science community.

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